

IT-IM 05

DATA WAREHOUSING AND DATA MINING

UNIT 1 - PART 1: DATA WAREHOUSE CONCEPTS

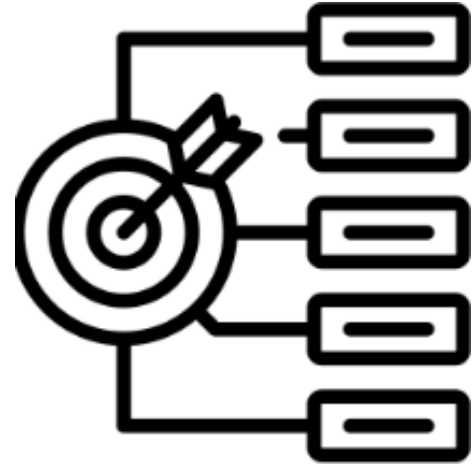
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SUBJECT TEACHER

Learning Objectives

At the end of the lesson, the students will be able to:

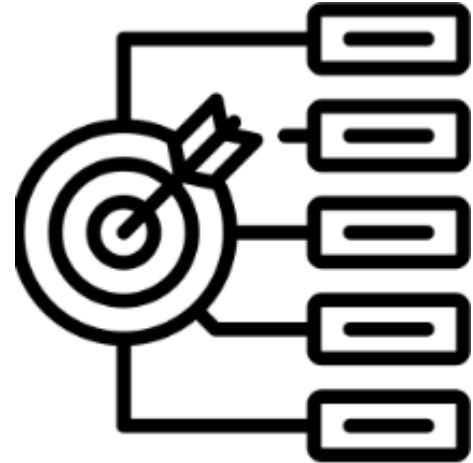
- describe the fundamental concepts of data warehousing by defining the term and identifying its purpose accuracy;
- differentiate a data warehouse, database, data mart, and data lake by correctly identifying their characteristics and appropriate applications;
- describe the advantages and limitations of data warehouses by relating their benefits and implementation challenges to the given specific scenarios;



Learning Objectives

At the end of the lesson, the students will be able to:

- explain the Extract, Transform, and Load (ETL) process by accurately describing each stage and its purpose;
- compare Online Transaction Processing (OLTP) and Online Analytical Processing (OLAP) by correctly selecting the appropriate technology for each scenario and provide valid justification.



Data Warehouse

- a repository that stores data from multiple sources that is organized, cleansed, and standardized for enterprise utilization.
- Helps organizations to derive business insights to improve decision-making.
- Typically stores current and historical data that is meant for analysis.



Characteristics

- Subject-oriented
 - Data is based on a theme or particular area.
 - Example: sales, product



Characteristics

- Integrated
 - Creates consistency on disparate data from various systems



Characteristics

- Time-variant
 - Data is organized in time intervals
 - Example: weekly, monthly, quarterly



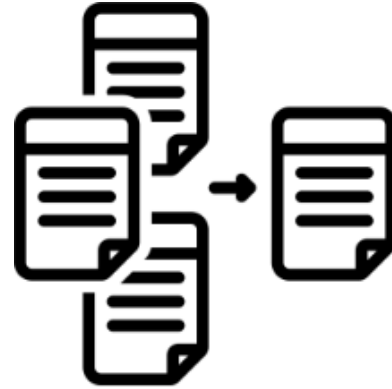
Characteristics

- Non-volatile
 - Once data is in the warehouse, it remains unchanged and stable (never deleted)

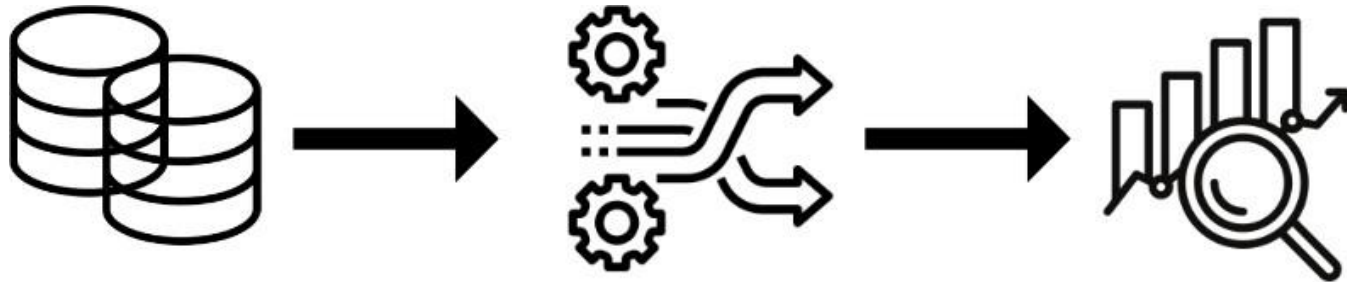


Characteristics

- Summarized
 - Data is often aggregated for optimized reporting



ETL Process



- **Extract:** Source data is extracted from operational or transactional systems
- **Transform:** Extracted data is transformed, formatted, corrected
- **Load:** Output is then loaded to the data warehouse for analytics and reporting

How is it different to a DB?

	Data warehouse	Database
Purpose	Analysis	Reporting
Database	OLAP (online analytical processing)	OLTP (online transactional processing)
Type of collection	Subject-oriented	Application-oriented
Query	Complex analytical queries	Simple transaction queries

Source: <https://www.coursera.org/articles/data-warehouse-vs-database>

OLAP

- **Online Analytical Processing**
- Data processing framework
- Supports decision-making
- Not time bound
- Allows complex queries
- Multiple records at a time
- Based on dimensional model
- Subject-oriented

OLTP

- **Online Transactional Processing**
- Data processing framework
- Supports day-to-day transactions
- Real-time
- Allows simple queries
- One record at a time
- Based on relational model
- Application-oriented

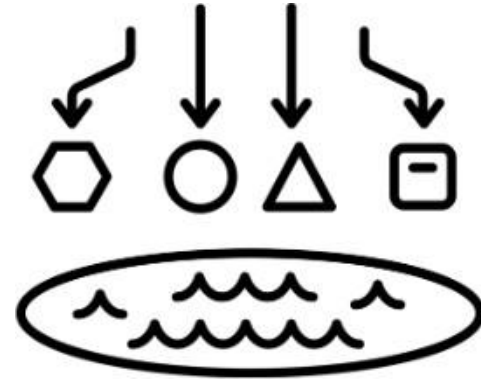
Data Marts

- These are subsets of a data warehouse
- If the data warehouse is for the whole enterprise, data marts are for departments
- Data is optimized for a specific department / unit.



Data Lakes

- Decentralized approach as opposed to the centralized approach of the data warehouse
- Includes **raw data** from operational and transactional sources unlike in a data warehouse where data has already been transformed.
- Usually follows ELT than ETL.



Scenarios

Example 1: E-commerce Site

Technology	Real-Life Example	Purpose
Database	When a customer places an order, updates their address, or makes a payment.	Supports daily business transactions (OLTP).
Data Warehouse	E-commerce store central analytics system that combines sales, inventory, logistics, and customer data from all regions.	Supports company-wide reporting and strategic decision-making.
Data Mart	The Marketing Department's dashboard showing campaign performance and customer purchasing trends.	Provides department-specific data for analysis.
Data Lake	Stores product images, customer reviews, chat messages, search history, clickstream data, and videos in their original format.	Stores raw structured and unstructured data for future analysis and machine learning.

Scenarios

Example 2: Streaming Service

Technology	Real-Life Example	Purpose
Database	Stores user accounts, subscriptions, watch history, and payments.	Supports day-to-day user transactions.
Data Warehouse	Consolidates global viewing statistics, subscriber growth, and revenue data.	Helps executives monitor business performance.
Data Mart	The Content Team analyzes which genres are trending to decide what shows to produce next.	Supports a specific department's analytics.
Data Lake	Stores raw video files, streaming logs, subtitles, clickstream data, and viewing behavior.	Stores massive amounts of raw data for AI and recommendation systems.

Advantages of DW

- Generates ROI (return on investments)
- High productivity may arise
- Assists in decision support such as in trend analysis, outcomes comparisons
- Can increase the operational value of the organization
- Enhanced customer service
- Easier access to historical data
- Enhance data quality and consistency



Disadvantages

- Requires a lot of resources
- Data ownership concerns
- Can be time-consuming and may result to having additional reports
- Necessary maintenance
- Complex integration



Problems / Limitations

- Underestimation of resources needed during ETL
- Hidden problems from source systems
- Required data may not be captured
- Homogenization of data that may cause information loss



Possible Deployments

- Traditionally, on-premises (inside the organization's local infrastructure)
- Nowadays, cloud can support data warehouses and the shift allows the company to focus more on the business goals since the responsibility for the upkeep is with a third party.



Big Data and NoSQL

- Big data is a concept that describes massive amounts of data (raw and unstructured) in various types and formats.
- NoSQL utilizes non-relational databases that can handle structured, semi-structured, and unstructured data.



Activity 1

- Research a company that implements data warehousing / data mart / or data lakes.
 1. Describe how they have implemented it. [5 points]
 2. What are the benefits gained from the implementation? [5 points]

Summarize your answers and please do not copy-paste directly from the Internet. Include complete sources.

Answers should be written in a one whole sheet of yellow paper.

References

- Bathia Parteek (2019). Data Mining and Data Warehousing. Cambridge University Press
- What is a Data Warehouse? Oracle. <https://www.oracle.com/ph/database/what-is-a-data-warehouse/>
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- Carrington, Tim. Data Warehousing and Business Intelligence. University of California, Irvine. Coursera. <https://www.coursera.org/learn/data-warehousing-business-intelligence/home/info>